GPU Rendering of Mandelbulb Fractals

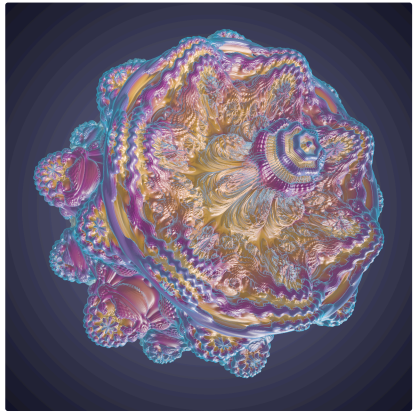
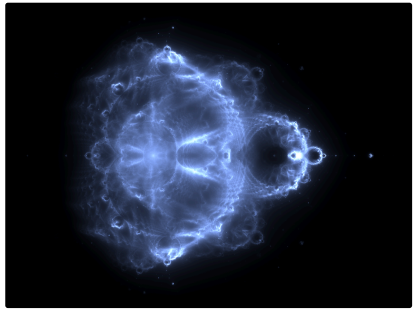
CS677 Final Project – Marcos Traverso, Joshua Hizgiaev

Goal

Render the **Mandelbulb**, a 3D fractal defined by iterated spherical-coordinate maps on the GPU with CUDA.

Where's the Parallelism

1. **Pixels independent**
   Each ray is its own CUDA thread; the screen grid maps directly onto the thread grid.
2. **ALU-heavy inner loop**
   The SDF evaluation is pure math with minimal warp divergence for neighboring pixels.

Features / Stretch Goals

- Volumetric fog
- Ambient occlusion
- Special coloring
- Interactive camera
- Alternative fractals/SDFs